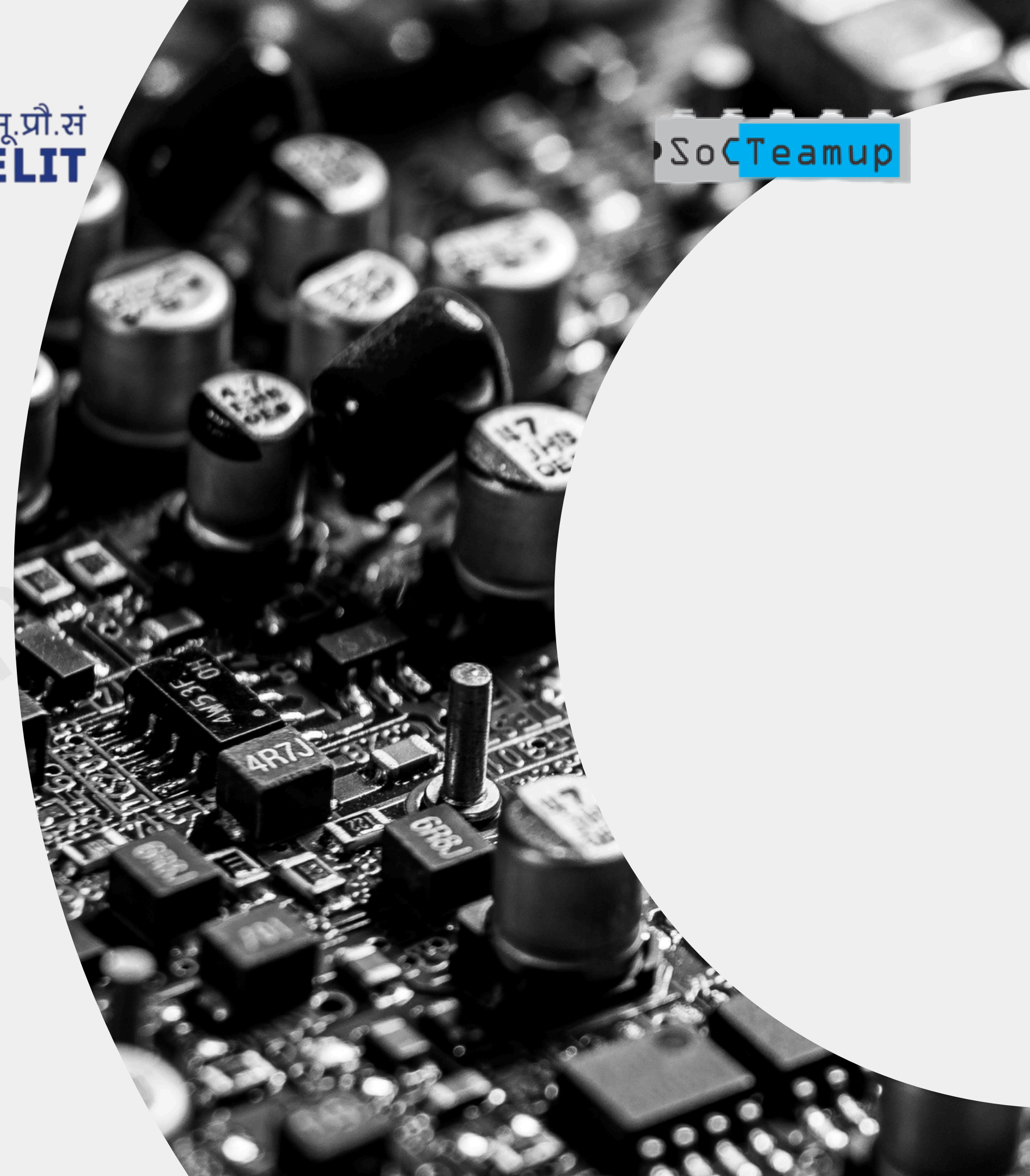


Static Timing Analysis (STA) Fundamentals



Outline

1	What is Static Timing Analysis (STA)?
2	Why STA is essential in digital design
3	Timing Paths: Launch to Capture
4	Setup and Hold Time Basics
5	Clock Edge and Timing Window
6	Types of Timing Paths
7	Combinational and Sequential Delays
8	Slack and Timing Margins
9	Clock Skew and Jitter
10	Single Clock Domain STA
11	Multi-Clock Domain STA
12	Common STA Constraints
13	Timing Reports and Interpretation
14	STA Tools Overview
15	Summary and Best Practices

➔ What is Static Timing Analysis (STA)?

STA is a method to verify whether timing requirements are met for a design.

It:

Analyzes all possible timing paths

Does not require simulation inputs

Works on design netlist and constraints

➔ What is **Static Timing Analysis (STA)**?

RTL is human-readable and
technology-independent

**Goal: Ensure signals arrive at the
right time neither too early nor
too late**

➔ Why STA is Essential in Digital Design

STA ensures:

- Timing correctness before tape-out.
- No setup/hold violations
- Reliable chip operation at target frequency
- Early detection of critical paths

➔ Why STA is Essential in Digital Design

Without STA:

- Silicon failure risk increases
- Cost of re-spin is very high

STA is mandatory in modern ASIC/SoC flows.

Timing Paths: Launch to Capture

A timing path is the path between:

- Launch flip-flop (start point)
- Combinational logic
- Capture flip-flop (end point)

Flow:

- Data launched at clock edge
- Propagates through logic
- Captured at next clock edge

Total path delay must satisfy timing constraints.

- Violations lead to incorrect latch behavior.
- Setup → Checked at slow corner
- Hold → Checked at fast corner

➔ Setup and Hold Time Basics

Setup Time:

- Minimum time data must be **stable before** the clock edge.
- $\text{Data arrival time} \leq \text{Required time}$

Hold Time:

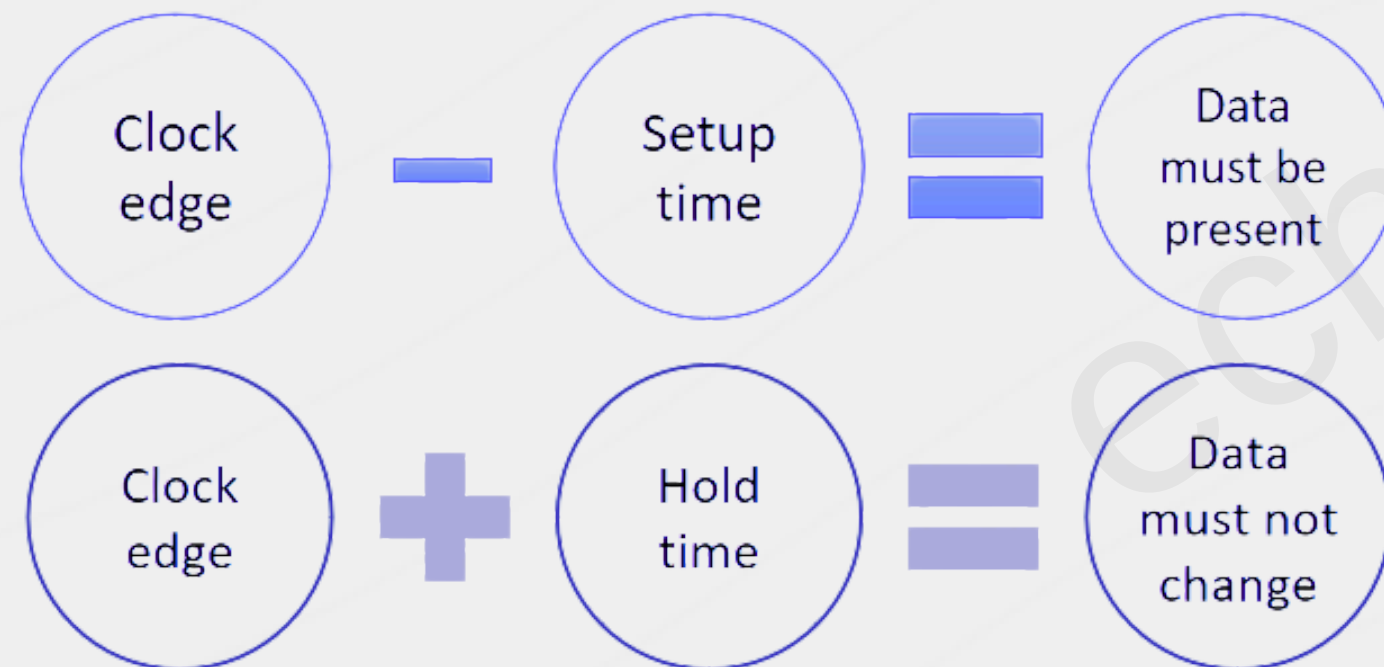
- Minimum time data must remain **stable after** the clock edge.
- $\text{Data arrival time} \geq \text{Hold requirement}$

Variation leads to incorrect behavior

- Setup must be checked at slow corner.
- Hold must be checked at fast corner.

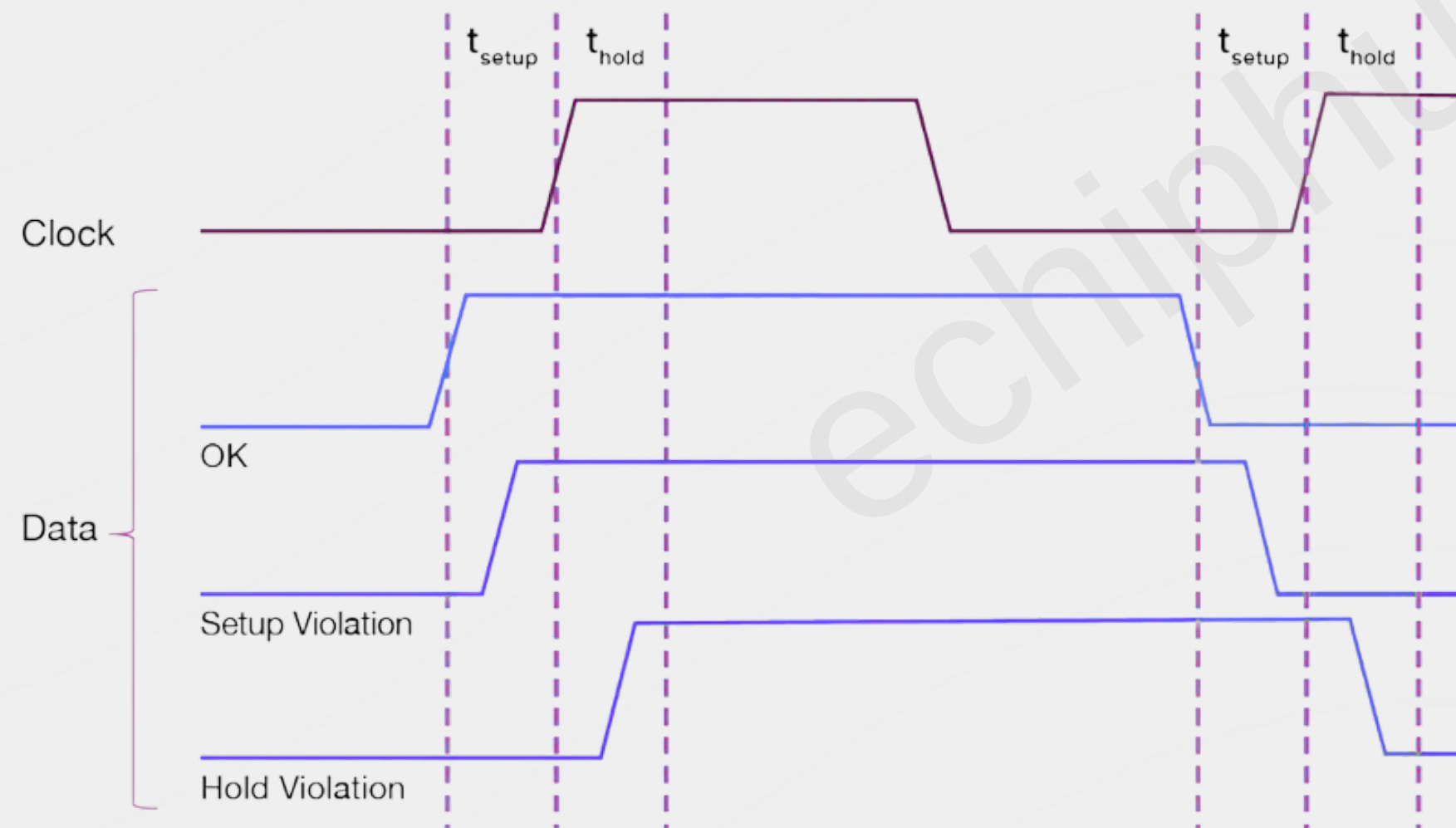
➔ Clock Edge and Timing Window

Data must arrive at the capturing register within window:



This defines the valid timing window for data propagation.

➔ Clock Edge and Timing Window (for setup and hold)



➔ Types of Timing Paths

- Launch → Capture Path
- Clock → Q → Comb Logic → D
- Input → Register (Input Path)
- Register → Output (Output Path)
- Clock Gating → Enable Path
- Asynchronous Paths (cross-domain)

STA checks all these path types individually.

Delays in Timing Paths



These delays are extracted
from .lib and .sdf files.

➔ Slack and Timing Margins



Positive Slack:

- Timing met

Negative Slack:

- Timing violation → needs fixing

➔ Clock Skew and Jitter

Clock Skew:

- Difference in clock arrival at source FF and destination FFs

Clock Jitter:

- Variability in clock period due to noise, PLL behavior

Both impact available slack and can cause timing failures.

➔ Single Clock Domain STA

Assumes all registers are triggered by the same clock.

Easier to analyze:

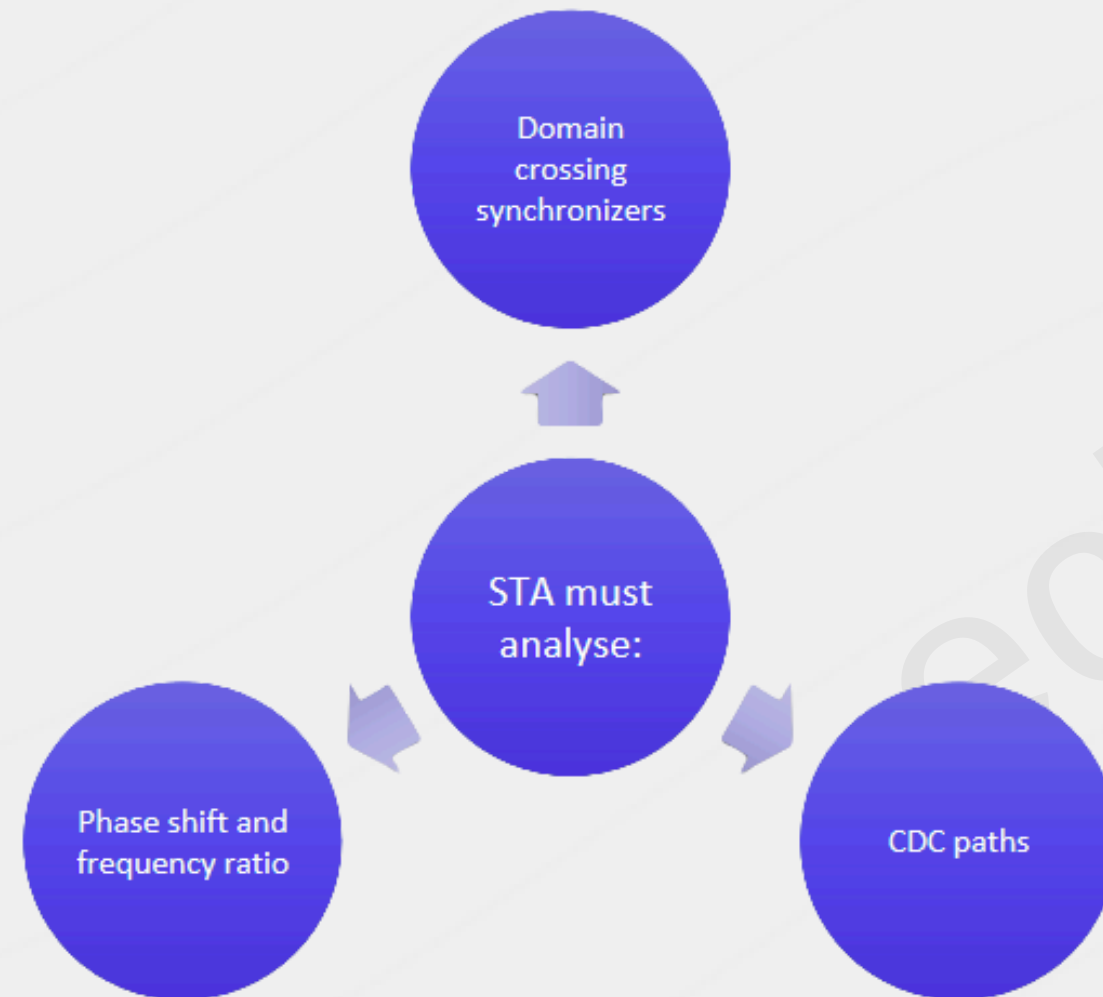
All launch/capture events on same
edge

Tools propagate timing from root
clock net

Useful for initial design sanity checks.

➔ Multi-Clock Domain STA

Different clock domains → Asynchronous interactions



Different clock domains → Asynchronous interactions

➔ Reading Timing Reports

Reports
show:

- Start point and End point
- Arrival and required times
- Path delay
- WNS and TNS
- Setup/Hold slack

Engineers fix critical paths based on worst slack reports.

➔ Timing Reports(OpenSTA)

```
OpenSTA> report_checks -path_delay max
Startpoint: sel (input port clocked by clk)
Endpoint: out (output port clocked by clk)
Path Group: clk
Path Type: max
```

Delay	Time	Description
0.00	0.00	clock clk (rise edge)
0.00	0.00	clock network delay (ideal)
5.00	5.00	v input external delay
0.00	5.00	v sel (in)
86.78	91.78	^ _4_/Y (OAI21X1)
0.00	91.78	^ out (out)
	91.78	data arrival time
100.00	100.00	clock clk (rise edge)
0.00	100.00	clock network delay (ideal)
0.00	100.00	clock reconvergence pessimism
-5.00	95.00	output external delay
	95.00	data required time
	95.00	data required time
	-91.78	data arrival time
	3.22	slack (MET)

➔ STA Tools Overview

Common STA Tools:

- Synopsys PrimeTime
- Cadence Tempus
- OpenSTA (open-source)
- Siemens Questa (timing-aware simulation)

STA is part of signoff flow before GDS export.

➔ Summary and Best Practices

Summary:

STA checks timing
without test vectors

Setup and Hold must be
verified across paths

Slack must be ≥ 0 for
reliable operation

Best Practices:

Always validate SDC
constraints

Check clock domains for
crossings

Analyze WNS and TNS
reports in detail

Thank You !

echiphu

